IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

OPTIMAL CONTROL

Friday 19 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer FOUR questions.

All questions carry equal marks

This is an OPEN BOOK exam

Any special instructions for invigilators and information for candidates are on page 1.

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OPTIMAL CONTROL

1. Find the equation of the curve $y(x) = (y_1(x), y_2(x))^T \in \mathbb{R}^2$ that is an extremal for the functional

$$J(y) = \int_0^{\frac{3\pi}{4}} \left((y_1(x))^2 + y_1'(x) y_2'(x) + (y_2'(x))^2 \right) dx$$

for the boundary conditions $y_1(0) = 2$, $y_1(\frac{3\pi}{4}) = 3$, $y_2(0) = 3$, and $y_2(\frac{3\pi}{4})$ is free. The functions $y_1(x)$ and $y_2(x)$ are independent.

(Hint: this is a two degree-of-freedom setting, and the same derivations for the necessary conditions are valid as for the single degree-of-freedom case, provided that L_y and $L_{y'}$ are interpreted as gradient vectors of L (where L is the Lagrangian), and products of vector quantities (such as $L_y \cdot \eta(x)$ and $L_{y'} \cdot \eta(\frac{3\pi}{4})$) are inner products in $\mathbb{R}^2$ (where $\eta(x) \in \mathbb{R}^2$ is the perturbation curve)).

a) Write the Lagrangian of the problem. [3]

b) Write the first-order necessary conditions for optimality. [10]

c) Compute the general form of the extremal curve. [4]

d) Calculate the extremal curve by using the boundary conditions. [3]

2. A scalar linear system is described by the differential equation

$$\dot{x}(t) = u(t) \qquad x(0) = 0$$

It is desired to find the control law that minimises the performance measure

$$J(u) = x(t_f)^2 + \int_0^{t_f} \left((x(t) - \cos t)^2 + u(t)^2 \right) dt$$

The admissible state and control values are not constrained.

a) Write the Hamilton-Jacobi-Bellman (HJB) differential equation, compute the control in terms of the value function that achieves the infimum in the HJB equation and simplify the HJB equation. [5]

b) Write the boundary condition that must be satisfied by the value function. [2]

c) Assume that the HJB differential equation admits a solution of the form

$$V(t,x) = K_1(t)x(t)^2 + 2K_2(t)x(t) + K_3(t)$$

where $K_1(t)$, $K_2(t)$, and $K_3(t)$ are unknown scalar functions of t.

i) Calculate $K_1(t_f)$, $K_2(t_f)$, and $K_3(t_f)$ so that the assumed solution matches the boundary condition found in part b). [2]

ii) Derive a system of ordinary differential equations for $K_1(t)$, $K_2(t)$, and $K_3(t)$, such that the value function $V(t,x)$ satisfies the HJB equation. [5]

iii) Calculate the optimal state trajectory as a function of time t. [4]

iv) Compute the optimal control law as a function of time t. [2]

3. A rocket, modelled as a particle of constant mass m moving in a zero gravity two-dimensional empty space, can be described by the following equations:

$$\begin{aligned}
\dot{x}_1(t) &= x_3(t) \\
\dot{x}_2(t) &= x_4(t) \\
\dot{x}_3(t) &= \frac{cu(t)}{m}\cos v(t) \\
\dot{x}_4(t) &= \frac{cu(t)}{m}\sin v(t)
\end{aligned}$$

in which the state $x_1(t)$ is the horizontal displacement, $x_2(t)$ is the altitude, $x_3(t)$ is the horizontal velocity, and $x_4(t)$ is the vertical velocity of the rocket. Also, $u(t) > 0\ \forall t \in [0, t_f]$ (where t_f is the final time) is the fuel mass flow rate, assumed to be a known function of time, c is a positive constant, $cu(t)$ is the total thrust, and $v(t)$ is the angle between the direction of the thrust and the horizontal positive direction. Consider the problem of finding the time minimising control $v(t)$ that takes the rocket from an initial position and velocity to a final position and velocity. The admissible controls are not bounded.

a) Write the functional $J(v)$ that should be minimised. [2]

b) Construct the Hamiltonian of the problem. [2]

c) By considering the maximisation of the Hamiltonian, derive the optimal control in terms of the states and costates (Hint: write the optimal control in terms of $\tan v^*(t)$) [5]

d) Derive the adjoint equations. [3]

e) Calculate the optimal control $\tan v^*(t)$ as a function of time and in terms of constant parameters. [4]

f) Instead of the time minimising problem, consider the problem of finding the control $v(t)$ minimising only a terminal cost $\phi(x(t_f))$ when some of the final states are free, where $x(t) = (x_1(t), x_2(t), x_3(t), x_4(t))^T \in \mathbb{R}^4$. Compute the optimal control as a function of time and discuss its differences from the optimal control calculated in part e). [4]

4. A manufacturer with production rate $x(t)$ at time t may allocate a fraction $u(t)$ of their production rate to reinvestments in a factory (thus increasing the production rate) and use the rest $(1-u(t))$ to store goods in a warehouse. Therefore, $x(t)$ evolves according to

$$\dot{x}(t) = \alpha u(t) x(t)$$

in which α is a given constant. The manufacturer would like to maximise the total amount of goods stored summed with the capacity of the factory at the given final time, t_f. This leads to the problem of finding $u(t)$ that minimises

$$-x(t_f) - \int_0^{t_f} (1-u(t)) x(t) dt$$

subject to

$$\begin{aligned} &\dot{x}(t) = \alpha u(t) x(t), \quad 0 < \alpha < 1 \\ &x(0) = x_0 > 0 \\ &0 \leq u(t) \leq 1, \ \ \forall t \in [0, t_f] \end{aligned}$$

a) Write the associated Hamilton-Jacobi-Bellman (HJB) differential equation. [2]

b) Compute the control that achieves the infimum in the HJB equation. [3]

c) Use the computed control to simplify the HJB equation. [2]

d) Write the boundary condition that must be satisfied by the value function. [2]

e) Assume that the HJB differential equation admits a C^1 solution of the form

$$V(t,x) = K(t) x(t)$$

where $K(t)$ is an unknown scalar function of t that is to be found.

i) Calculate $K(t_f)$ so that the assumed solution matches the boundary condition found in part d). [2]

ii) Derive and solve a differential equation for $K(t)$. [7]

iii) Compute the optimal control law as a function of time. [2]